



Uncovering patterns of multichannel mobile communications

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ABSTRACT

Human communication patterns can be discovered or examined by analyzing mobile phone data. However, previous studies have mainly concentrated on analyzing one type of data, such as calling or Short Messaging Service (SMS). This limited approach hinders our comprehensive understanding of human communication behaviors as a whole. In this paper, we combine large-scale data and entropy evaluation methods to examine the patterns of multichannel mobile communications from various perspectives. First, we showcase users' preferences across various channels and reveal that there is a U-shaped interactive relationship between calling and SMS in dual-channel communication as the number of contacts increases. Second, we identify an upper limit for communication vitality in multichannel communications, with the corresponding number of contacts being 10, 100, and 20 in calling, SMS, and hybrid channel. Furthermore, the patterns of strength allocation and communication vitality for both SMS and calling by users are very comparable. Third, compared to calling, users tend to send text messages more regularly and at shorter intervals. And they usually have a shorter delay between texting and calling than from calling to texting. Finally, we propose a method to quantify the complexity of multichannel behavior sequences based on multiscale entropy and two null models. These findings can facilitate a deeper understanding of human communication behaviors, while our novel method offers a universal framework for quantifying the complexity of multichannel behaviors.

1. Introduction

With the rapid development of the mobile communication industry, mobile phones have emerged as the primary means for daily human communication [1], leading to an explosion of mobile phone data. As the principal source reflecting human behavior and mobility patterns [2,3], mobile phone data has always been a subject of interest for researchers. Over time, researchers have extensively analyzed mobile phone data to study various aspects, including power exponent, communication time patterns, social fingerprint, and contact prediction. They have also applied this data to human social networks, such as addressing city problems [4,5], inferring social structures [6], studying social economy [7,8], and understanding the transport system [9]. Despite these numerous studies, most of them only focused on a single channel, either calling or Short Messaging Service (SMS), lacking studies on multichannel data.

In recent years, there has been a shift towards analyzing multichannel mobile phone data to gain a comprehensive understanding of human behavior and interactions. Heydari et al. found that there is less overlap between calling and SMS, and individual social

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signatures are consistent across different communication channels [10]. Li et al. found that communities in SMS networks are sparser and more unstable than those in calling networks [11]. Hayat et al. proposed a prediction model based on dual-channel data, which could be used to predict users' contacts and contact channels with high accuracy [12]. In spite of recent efforts to study multichannel data, the researchers overlooked the complexity of multichannel communications, which is an essential aspect of understanding this type of communication.

In this paper, we combine large-scale data and entropy evaluation methods to examine the complexity of multichannel mobile communications from various perspectives. First, we study users' preferences in multichannel communications. We examine the relationship between communication channels and social circle size (i.e. the number of contacts a user has) and find a U-shaped pattern between the usage of calling and SMS in dual-channel communication. Second, we study the communication vitality and the strength allocation patterns of the three channels. Third, we study multichannel mobile communications on the temporal scale. We propose the intraday patterns of three channels. We use actual entropy to measure the regularity of multichannel communications on the time scale. Statistical properties of multichannel communication behaviors based on intervals are presented. Finally, we propose a novel framework to quantify the complexity of the behavior sequence of multi-channel behaviors.

2. Results

2.1. Data description

Our mobile phone dataset consists of calling and SMS data; all phone numbers have been encrypted to protect privacy. The dataset covers the period from 1st Feb 2012 to 9th Feb 2012. The calling data includes 22,740,000 records and 550,000 users. The main items contain user ID, call start time, call end time, and the duration of every call. The SMS data contains 6,490,000 records and 280,000 users. The main items contain user ID, and sending time for each short message. The number of users who used dual-channel communications is about 270,000.

2.2. Behavioral preferences in multichannel communications

This study begins by examining users' preferences in multichannel communications. In communication activities, everyone has a social circle that consists of his/her contacts. The size of a social circle, defined as the total number of contacts, can be used to study communication patterns. To this end, we compute the total number of contacts of each user. When users use multichannel mobile communications, their contacts can be divided into four types: calling contacts (received calls), SMS contacts (received text messages), dual-channel contacts (received both calls and text messages), and single-channel contacts (received either a call or a text message). Fig. 1 shows the relationship between different types of contacts and the size of social circles.

A comparison of calling contacts and SMS contacts shows that the number of SMS contacts increases with the size of social circles, while the number of calling contacts increases first and then plummets when the number reaches 200; there are always more calling contacts than SMS contacts before reaching this threshold. We also find that the number of dual-channel contacts first keeps increasing, and then is followed by a slight decrease when the total contacts number is around 500. The reason for this phenomenon is that at a social circle size of 500, far more people choose SMS than calling. In addition, the number of single-channel contacts highly overlaps with calling contacts and consistently rises as the social circle expands; it is also always more than dual-channel contacts. This indicates that compared with dual-channel communication, most users prefer single-channel communication, with calling being the most common choice.

All of the above behavioral preferences arise from the difference between the two modes of communication, i.e., calls are more convenient for delivering a large amount of information to a small number of contacts, while SMS is more efficient when many people need to be contacted at the same time.

To gain a more detailed understanding of the interactive usage between calling and SMS in dual-channel communications, we calculate the ratio of the number of SMS contacts to the number of total contacts, which is denoted as R_{SMS} . As shown in Fig. 2, when considering the total number of contacts, there is a U-shaped relationship between calling and SMS in dual-channel communication. Specifically, when the number of total contacts is less than 40, the number of users preferring SMS will gradually decrease, while when the number is more than 40, the number of users preferring SMS tends to increase. It shows that the trend of users' communication behavior is non-monotonic as the number of total contacts changes. This pattern may arise from the distinction between the two communication mechanisms: users can send messages to multiple contacts simultaneously, whereas they can only make calls to individuals one by one. Therefore, when the number of total contacts is large, users tend to use SMS.

2.3. Communication strength in multichannel communications

In this section, we further explore the frequency of users communicating with their contacts across each channel, which can be used to indicate a user's communication vitality. We define communication strength as the number of communication records between a user and their contact. We observe the average strength of every user, that is, the average number of communication records that a user assigned to each of their contacts. As shown in Fig. 3, the average strength in calling, SMS, and hybrid data tends to increase to a threshold and then decrease. This threshold is the number of contacts corresponding to the highest point of the average strength. When the number of contacts reaches the threshold, the average social strength begins to decline. It reminds us of Dunbar's theory that human cognition limits the capacity of social circles, allowing humans to have a stable number of contacts

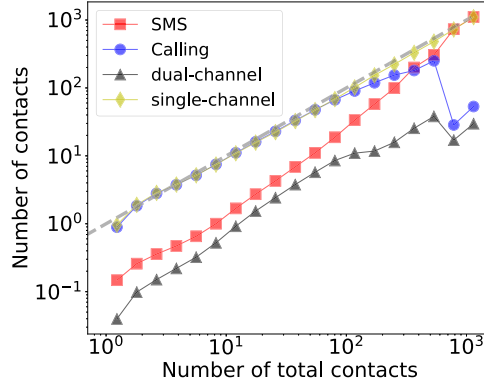


Fig. 1. Relationships between communication channels and social circle size. SMS refers to contacts who received text messages, while Calling refers to contacts who received calls. Dual-channel refers to contacts who received both calls and text messages, while single-channel refers to contacts who received either a call or a text message. The number of total contacts represents the size of social circles.

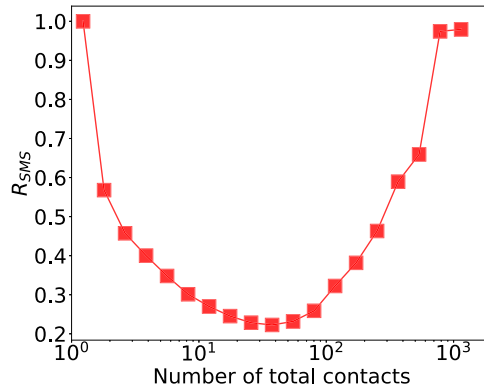


Fig. 2. The relationship between SMS and calling in dual-channel communication. R_{SMS} is the ratio of the number of SMS contacts to the number of total contacts.

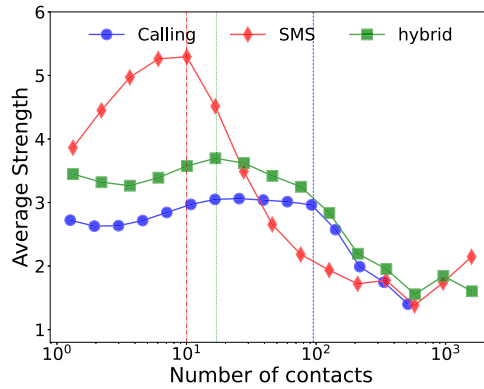


Fig. 3. The communication vitality in multichannel communications. The average strength is the average communication frequency between an individual and his or her contacts. It is a measure of a user's communication vitality. The hybrid data is defined as the combination of calling and SMS data.

which is 148 [13]. However, the thresholds vary significantly across communication channels: 10 for SMS, 100 for calling, and an intermediate threshold (20) for hybrid data. Due to the count of calling records being more than that of SMS, the threshold for hybrid data are closer to that for calling.

To further study the communication strength, we then provide the strength distributions of the calling, SMS, and hybrid data. As shown in Fig. 4(a), all these distributions follow the power-law distribution: larger strength has a lower probability of occurring, while smaller strength has a higher probability of occurring. However, the distribution scale is different for each communication

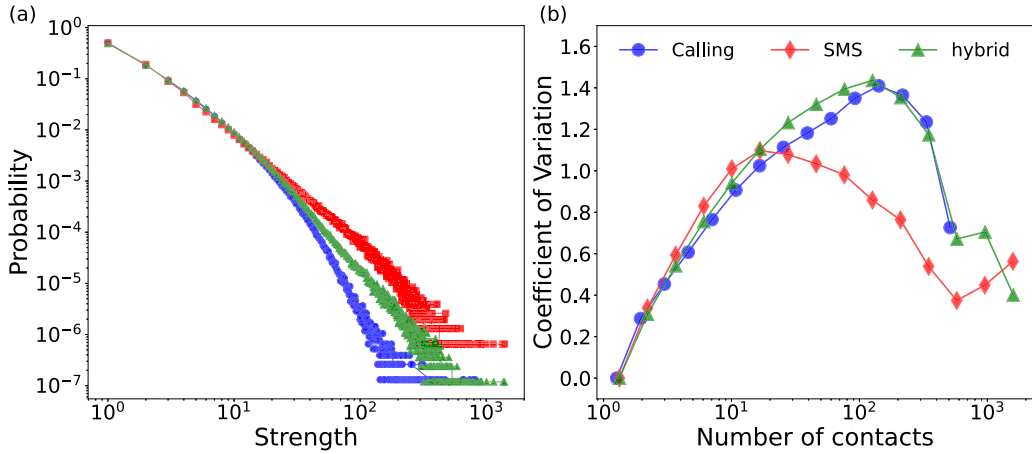


Fig. 4. The strength distributions and individual strength distribution patterns in multichannel communications. (a) Strength distributions of calling, SMS and hybrid data. (b) The CV of individuals' strength sequences of calling, SMS and hybrid data.

channel, with calling having a larger probability scale than SMS. This suggests that the difference in strength is more pronounced in calling than in SMS.

Is there a difference in individual strength distributions in multichannel communications? To answer this question, we analyze the dispersion degree of each user's strength sequence. The strength sequence of a user consists of the strength of each of his/her contacts. We compute the coefficient of variation (CV) of these strength sequences to measure their dispersion degree; the larger the CV is, the higher the dispersion degree is, and vice versa. As shown in Fig. 4(b), CVs increase first and then decrease with the number of contacts, and the peak values of SMS and calling are close to 10 and 100, respectively. Both the overall trend and peak value are similar to that of the average strength shown in Fig. 3. The results reveal a pattern in human communications: when a user's social energy (the average strength) is closer to the peak, their strength sequence tends to be more discrete, indicating that they tend to contact only a few people with disproportionately high frequency. However, when the number of a user's contacts exceeds this peak, this pattern fades, and the strength sequence with their contacts becomes more even, suggesting that they tend to contact a larger number of friends at a similar frequency.

2.4. Temporal regularity of multichannel communications

In the above sections, we have studied how the size of a user's social circle relates to their communication patterns. In this section, we will focus on the temporal regularity of multichannel communications. We start by examining the intraday pattern [14,15] for calling, SMS and the hybrid data. As shown in Fig. 5, the intraday pattern of both calling and SMS usage follows the human circadian rhythm, meaning that people tend to use both communications more during the day and less at night. However, there are still significant differences in the specific times when people use calling and SMS. To be specific, the intraday pattern of calling is bimodal, with peaks at 10:00 and 17:00, while the most frequent use of SMS is at 21:00. This reveals that humans prefer to call during working hours while sending short messages when approaching bedtime. Besides, the intraday pattern of hybrid data is identical to that of calling, which may be because the number of calling records is more than SMS.

In order to quantify the regularity of various communications, we extract each user's time series and compute their actual entropy: the smaller the actual entropy, the more robust regularity of the time series. As shown in Fig. 6, the distributions of actual entropy in calling, SMS and hybrid data follow a unimodal distribution, and the actual entropy corresponding to the peak of calling is the highest among the three communication channels. This indicates that the timing of calls is less regular than the timing of SMS or hybrid data. Specifically, the actual entropy distributions of calling and SMS are distinct in two different ranges: SMS is most distributed in the range with lower actual entropy ($[0,2]$), while that of calling is concentrated in the range with larger actual entropy ($[2,3]$). Hence, the regularity of SMS is more potent than that of calling. The distribution of the hybrid data is between that of calling and SMS and tends to resemble that of calling. This is because there are more records of calling than SMS.

Do users with stronger regularity exhibit the same intraday pattern? To answer this question, we select the top 5% of users from the SMS data—those with the lowest actual entropy and thus the strongest regularity. Next, we calculate the intraday pattern of each user and cluster these patterns using the k-means algorithm (we use the k-means classifier from the sklearn module in Python and set the number of categories to 15). We select five categories of intraday patterns for visualization in Fig. 7(a)–(e). Within each subfigure, the thin lines represent the intraday patterns of individual users within the respective category, while the thick line represents the average pattern for that category. The results reveal that users with stronger regularity exhibit diverse intraday patterns, with messaging times spanning from morning to night. However, they all tend to concentrate their communication activities during a particular period. To further validate this finding, we also select five users with the highest actual entropy. As illustrated in

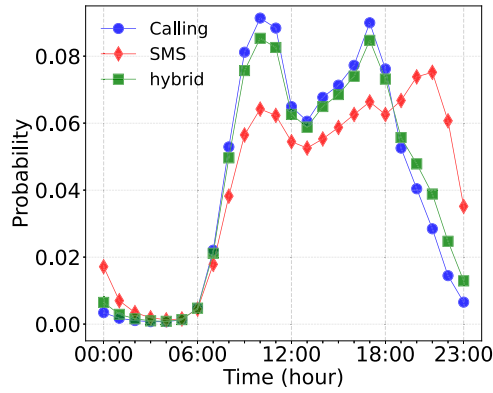


Fig. 5. The intraday pattern of calling, SMS and hybrid data. The intraday pattern is a probability distribution of the number of recorded behavior in hours during a day. We calculate the intraday pattern of calling, SMS and hybrid data, respectively.

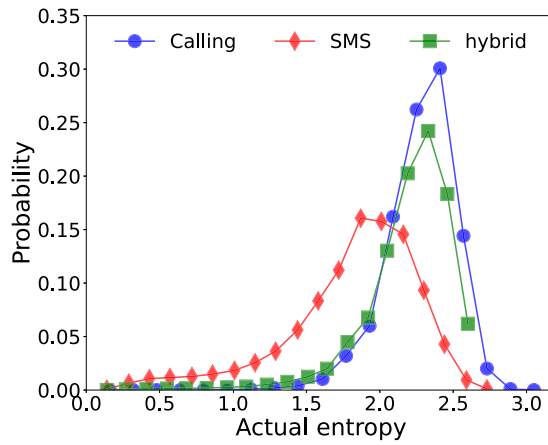


Fig. 6. The distributions of actual entropy in calling, SMS and hybrid data.

Fig. 7(f), the intraday patterns of these users are scattered. This confirms that the actual entropy can accurately reflect the regularity of users' communication time series.

In addition to communication timing, the time interval is often used as one of the important statistical properties in dynamic research. To study the interval distribution of multichannel communications, we calculate the intervals for three channels respectively. This entails measuring the time difference between the start times of two consecutive records, and the resulting distributions are shown in Fig. 8(a). It can be observed that when intervals are smaller, SMS data occupies a large proportion, while when intervals are larger, calling data takes up a higher percentage. It can be concluded that the intervals of SMS are generally smaller than that of calling.

In addition to studying the above time interval distribution of different communication channels independently, we further pay attention to the intervals between dual-channel communication. These intervals can be divided into two types: calling to SMS and SMS to calling. As shown in Fig. 8(b), when intervals are shorter, SMS to calling holds a higher percentage, whereas for longer intervals, calling to SMS becomes more predominant. This indicates that users tend to initiate a call shortly after sending a short message.

2.5. Quantifying the complexity of multichannel communications

In the above sections, we have analyzed users' communication patterns and revealed potential regularities in the utilization of multichannel communications based on preference and time. In this section, we will propose a novel framework to quantify the complexity of these multichannel communication behaviors. The framework is based on the behavior-shuffling null model and the behavior-ordering null model (see Methods). Specifically, we first extract the sequence of users' communication behaviors at the temporal scale. Then we construct two null models of the original sequence: the behavior-shuffling null model, which is the most complicated copy, and the behavior-ordering null model, which is the most ordered copy. The two null models only change the order of the communication behavior, without changing behavior types and counts. Finally, we take a proportion p to shuffle the

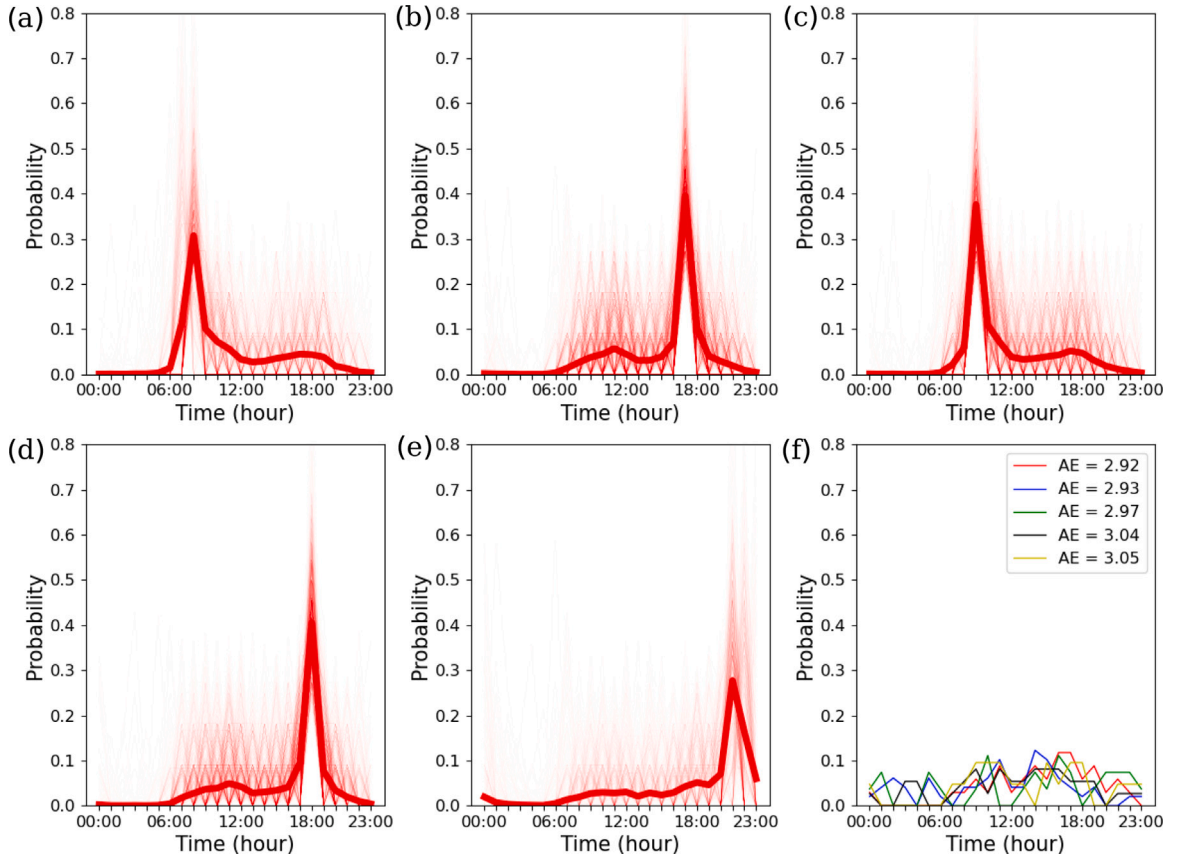


Fig. 7. The intraday patterns of users with widely varied actual entropy, respectively. (a)–(e) The intraday patterns of 5% of users with the lowest actual entropy; (f) The intraday patterns of five users with the highest actual entropy.

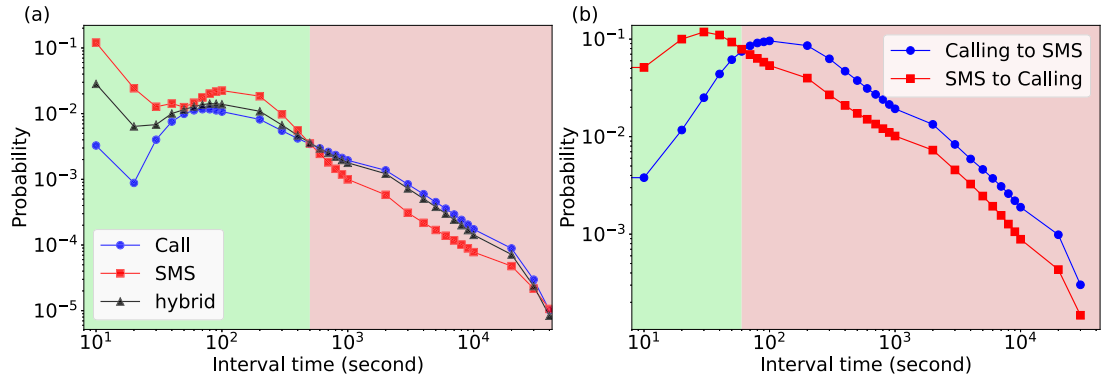


Fig. 8. The intervals of multichannel communications. (a) The distribution of the intervals of calling, SMS and hybrid data. (b) The distribution of the intervals of calling to SMS and SMS to calling.

sequence of the behavior-ordering null model, and if its multiscale entropy (MSE) (see Methods) is approximated to that of the original sequence at each time scale, the complexity of the original sequence is p (p ranges between 0 and 1, and the larger the p , the stronger the complexity).

As shown in Fig. 9, we calculate the MSE of the original behavior sequence and its two null models. The entropy value corresponding to each scale is obtained by averaging the results over 100 iterations. The entropy of the original sequence results increases significantly as the scale factor rises from 1 to 3. It shows that the self-similarity of the original sequence is relatively

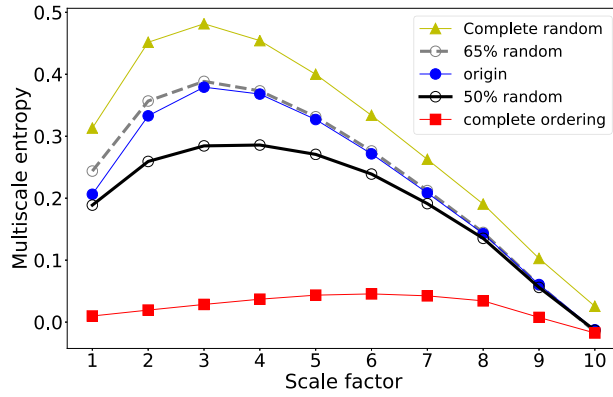


Fig. 9. Quantifying the complexity of multichannel communications The blue circle solid line represents the MSE of the original communication behavior sequence. The red rectangle solid line represents the MSE of the complete ordering null model. The black circle solid line and gray circle dotted line denote the MSE of the 50% random null model and 65% random null model, respectively. The yellow triangle solid line represents the MSE of the complete random null model. The correlation between the MES of the 65% shuffled behavior-ordering null model and the original data is 0.995.

low, and there is a certain level of complexity in the sequence. By comparing the results of the two null models with that of the original sequence, it can be found that the behavior-shuffling null model has the highest entropy at these scales, indicating that the complexity of this null model is also the strongest. On the other hand, the behavior-ordering null model displays the lowest entropy at each scale, which means that its self-similarity is the largest, and thus the complexity is the lowest.

The entropy of the original sequence falls between that of the two null models, indicating that its complexity is between 0 and 1. We consider the stopping condition to be that the multiscale entropy of the null-model sequence is sufficiently close to the multiscale entropy of the original sequence. To determine this stopping condition, we employ the Pearson correlation coefficient to calculate the multiscale entropy values of both types of sequences. If this correlation coefficient reaches 0.9 or higher, we consider it as a suitable termination condition. To determine this stopping condition, we employ the Pearson correlation coefficient to calculate the multiscale entropy values of both types of sequences. If this correlation coefficient reaches 0.9 or higher, we consider it as a suitable termination condition. According to the framework proposed above for quantifying the complexity of the communication behavior sequence, we choose to shuffle 50% and 65% of the behavior-ordering sequence. The results show that when the shuffling proportion is 65%, the MSE of the shuffled sequence is closed to that of the original sequence. The correlation between the MES of the 65% shuffled behavior-ordering null model and the original data is 0.995. Hence, the quantized value of the complexity of the original sequence is 0.65.

3. Materials and method

3.1. Actual entropy

Actual entropy can be used to quantify the regularity of any sequence [16,17], which is defined as

$$S_{\mathcal{E}} = \left(\frac{1}{n} \sum_{i=1}^n \Lambda_i \right)^{-1} \ln n, \quad (1)$$

where Λ_i represents the length of a never-before-seen shortest subsequence starting at position i in sequence $\mathcal{E}$. If there is no shortest subsequence, set $\Lambda_i = n - i + 2$ [17]. The smaller the value of actual entropy, the more ordered the sequence. Here, we choose to calculate the actual entropy, rather than the Shannon entropy [18] or Simpson's diversity index [19], because we aim to measure not only the concentration of sequences, but also to measure whether they are in order.

The timestamp format for calling and SMS data is composed of *the date* and *the time*. The date represents a specific day, accurate to one day, while the time represents a specific time within a day, accurate to one second. After extracting these time components from the timestamp format, the data is first sorted by date, and then by the time of each day. In the regularity analysis, we focus on *the time* within a day instead of *the date*, so we extract the specific time points after ordering as a time series. For example, first, we extract the time components of a user's call time as {22:33 2-1, 21:48 2-3, 18:40 2-4, 21:05 2-1, 21:13 2-2}. Then, we sort it by date and time as {21:05 2-1, 22:33 2-1, 21:13 2-2, 21:48 2-3, 18:40 2-4}. Finally, we extract *the time* in it as a time series [21:05, 22:33, 21:13, 21:48, 18:40]. When we process the time series, the time of a day is divided into 48 segments according to half hour as a segment (the first segment is 0:01-0:30, the second is 0:31-1:00...and so on), and coded as 1-48. Thus, the previous time series $\{t_1, t_2, \dots, t_n\}$ can be converted to $\{t'_1, t'_2, \dots, t'_n\}$ ($t'_i \in \{1, 2, \dots, 48\}$). For example, if a user's call time sequence is [21:05, 22:33, 21:13, 21:48, 18:40], it can be converted to $\mathcal{E} = \{43, 46, 43, 44, 38\}$ according to the above rule, and the actual entropy is 1.34.

3.2. Multiscale entropy

Multiscale entropy (MSE), first proposed by Costa et al. has been extensively used in analyzing physiological time series [20,21]. This method is based on sample entropy and is employed to quantify the complexity of time series across varying scales. It involves three key parameters, namely τ , m , and r . Here, τ represents the scale factor, m refers to the dimension of the embedded vector, while r denotes the threshold or similarity coefficient. The computation of multiscale entropy typically entails the following procedure.

Step 1: To set up a discrete time series, we start by defining a sequence of points $x_1, x_2, \dots, x_L$, where L denotes the number of points. The new time series can be obtained through a coarse-grained transformation, which is expressed mathematically as follows:

$$y_j^{(\tau)} = \frac{1}{\tau} \sum_{i=(j-1)\tau+1}^{j\tau} x(i), \quad (2)$$

where τ is the scale factor, and the length of each coarse-grained time series is $\frac{L}{\tau}$. When τ is equal to one, the coarse-grained time series is the same as the original time series. If τ is larger, it means that the time series is coarsely granulated.

Step 2: The time series is acquired by observing changes in scale denoted as τ , with a length of N where N equals L divided by τ . A collection of m -dimensional vectors is formed by taking consecutive serial numbers from $\mathbf{Y}^{(\tau)}(1)$ to $\mathbf{Y}^{(\tau)}(N - m + 1)$,

$$\mathbf{Y}^{(\tau)}(i) = [y^{(\tau)}(i), y^{(\tau)}(i + 1), \dots, y^{(\tau)}(i + m - 1)], \quad (3)$$

where these vectors represent m consecutive y values starting at the point i , with the range of i being $[1, N - m + 1]$. **Step 3:** $d[\mathbf{Y}^{(\tau)}(i), \mathbf{Y}^{(\tau)}(j)]$ can be defined as the maximum value of the vector $\mathbf{Y}^{(\tau)}(i)$ and $\mathbf{Y}^{(\tau)}(j)$ on the scale τ when the corresponding elements meet and take the absolute value, namely the following equation:

$$d[\mathbf{Y}^{(\tau)}(i), \mathbf{Y}^{(\tau)}(j)] = \max |y^{(\tau)}(i + k) - y^{(\tau)}(j + k)|, \quad (4)$$

where the absolute value of the difference between the corresponding elements in $\mathbf{Y}^{(\tau)}(i)$ and $\mathbf{Y}^{(\tau)}(j)$ is less than r . Next, for each value of i , calculate the distance $d[\mathbf{Y}^{(\tau)}(i), \mathbf{Y}^{(\tau)}(j)]$ between $\mathbf{Y}^{(\tau)}(i)$ and the rest of the vector $\mathbf{Y}^{(\tau)}(j)$, the range of k is $[0, m - 1]$, and the range of i and j are both $[1, N - m + 1]$, $i \neq j$.

Step 4: For a specified threshold value, denoted as r , we count the number of instances where $d[\mathbf{Y}^{(\tau)}(i), \mathbf{Y}^{(\tau)}(j)]$ is less than r , for all i such that $i \leq N - m + 1$. The ratio of this count to the total number of distances $N - m$, denoted as $C_i^{\tau, m}(r)$, is

$$C_i^{\tau, m}(r) = \frac{1}{N - m} |\{(i, j) | d[\mathbf{Y}^{(\tau)}(i), \mathbf{Y}^{(\tau)}(j)] < r\}|, \quad (5)$$

where the ranges of i and j are both $[1, N - m + 1]$, $i \neq j$.

Step 5: All points are averaged, as shown in the following formula:

$$C^{\tau, m}(r) = (N - M + 1)^{-1} \sum_{i=1}^{N-m+1} C_i^{\tau, m}(r). \quad (6)$$

Step 6: The dimension is increased to $m + 1$, repeat steps 2 to 5, and $C_i^{\tau, m+1}(r)$ is obtained on the scale τ and under the dimension $m + 1$. Calculate the average of $C_i^{\tau, m+1}(r)$ and get $C^{\tau, m+1}(r)$. Theoretically, the sample entropy value of the time series on scale τ is defined as

$$\text{SampEn}(m, r) = \lim_{N \rightarrow \infty} -\ln [C^{\tau, m+1}(r) / C^{\tau, m}(r)]. \quad (7)$$

When N is a finite value, the sample entropy of the sequence on the scale τ is obtained by the above steps, and the estimated value is recorded as

$$\text{SampEn}(\tau, m, r) = -\ln [C^{\tau, m+1}(r) / C^{\tau, m}(r)]. \quad (8)$$

Multiscale entropy is defined as a set of sample entropy values at multiple scales, so the multiscale entropy value is

$$MSE = \{\text{SampEn}(\tau, m, r) \mid \tau \in N^*\}. \quad (9)$$

Sample entropy is a useful tool to measure the degree of irregularity of a sequence on a single scale. The higher the self-similarity of the sequence, the larger the sample entropy value, the more complex the sequence. Similarly, multiscale entropy calculates the sample entropy of time series on multiple scales, providing further insight into the complexity of a sequence. Specifically, if the entropy value of the sequence monotonically decreases on the scale, the sequence has low self-similarity, a simple structure, and belongs to a random sequence. Conversely, if the entropy value is larger on the scale, the sequence's self-similarity is large, and its complexity is significant as well. Comparing the entropy of two sequences, if one has a higher entropy value on most scales, then that sequence is more complicated than the other.

3.3. Two null models for multichannel communications

The null model can serve as a reference for the original sample, as it is randomized while maintaining certain characteristics of the sample. It is helpful to quantify abstract contrasting relationships or properties in concrete terms [22,23]. Here we define two null models that can adjust the order and complexity of communication sequences, which can provide a reference and basis for measuring the cooperation of user communication sequences. In order to facilitate future calculations, the communication sequence is first symbolized: we define the user's calling behavior as 1 and SMS as 2. Then, the communication sequence is ordered by time, and the final symbolized sequence can be obtained. For example, if a user's communication record is {call, 10:23; SMS, 11:19; SMS, 17:05; call, 21:03; SMS, 23:12}, the symbolized communication behavior sequence of this user is [1,2,2,1,2].

The first one is the behavior-ordering null model, which is used to order the original communication behavior sequence in ascending order. For example, if the original sequence is [1,2,2,1,2], its behavior-ordering null model is [1,1,2,2,2]. The behavior-ordering null model preserves the behavior types in the original communication behavior sequence and sorts the order of these behaviors according to when they occurred, which can maximize the orderliness of the original sequence.

The other null model is the behavior-shuffling null model. It can adjust the complexity of the communication behavior sequence to the maximum. Take a simple communication sequence [1,2,2,1,2] as an example. By randomizing this sequence, its behavior-shuffling null model can be constructed. The behavior-shuffling null model preserves the type of communication behavior in the original sequence while shuffles the order of these behaviors, which can minimize the orderliness of the original sequence.

4. Conclusion

In summary, we explore the complexity of multichannel communications from various perspectives based on the large-scale data and information physics methodology. To be specific, we investigate the patterns in the usage of different communication channels (such as calling and SMS) when the social circle size changes. Furthermore, we examine the communication vitality of multiple channels. In addition, we examine their temporal regularities, including the intraday pattern and the regularity of communication time series. In the end, we propose a framework based on actual entropy and two null models to quantify the complexity of the multichannel communications sequence. We find that users are found to use multi-channel communication less frequently than they used a single channel. In the case of dual-channel communication, there is a U-shaped interactive relationship between calling and SMS as the total number of contacts changes. As the number of contacts is approaching 40, users gradually increase their utilization of calls; beyond that point, users start to increase their use of text messages as the number continues to rise.

In addition, we uncover that there is an upper limit to the number of contacts that users can effectively communicate with for each channel communication, which is defined as communication vitality. And there is a significant difference between the upper limit of different channels, with SMS being 10, and calling being 100. To our surprise, the limit has an effect on the communication pattern. That is to say, when the number of users' contacts is less than the limit, they tend to interact with only a few contacts frequently; while if it were the other way around, the user tends to balance their communication strength with more contacts.

Multichannel communications also have some specific patterns at the time scale. By studying intraday patterns, we find that users prefer to make phone calls during working hours while being willing to use SMS in the evening. And users send text messages more regularly than make calls. Besides, it can be observed that the intervals between two SMS activities are shorter than those of calling. In dual-channel communication, users are inclined to make phone calls shortly after sending text messages, in contrast to texting after making calls. Finally, when quantifying the complexity of multichannel behavior sequences using MSE, we ascertain that this dataset exhibits a certain degree of complexity, with a quantitative value of 0.65 representing the level of complexity.

These empirical findings reveal the relations and patterns within multichannel mobile communications from various perspectives, contributing to a deeper comprehension of social networks and human dynamics. Moreover, they hold significant guidance for mobile phone operators to achieve substantial gains.

CRediT authorship contribution statement

Bo Liu: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Xiao-Ke Xu:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Linyuan Lü:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

Data will be made available on request.

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